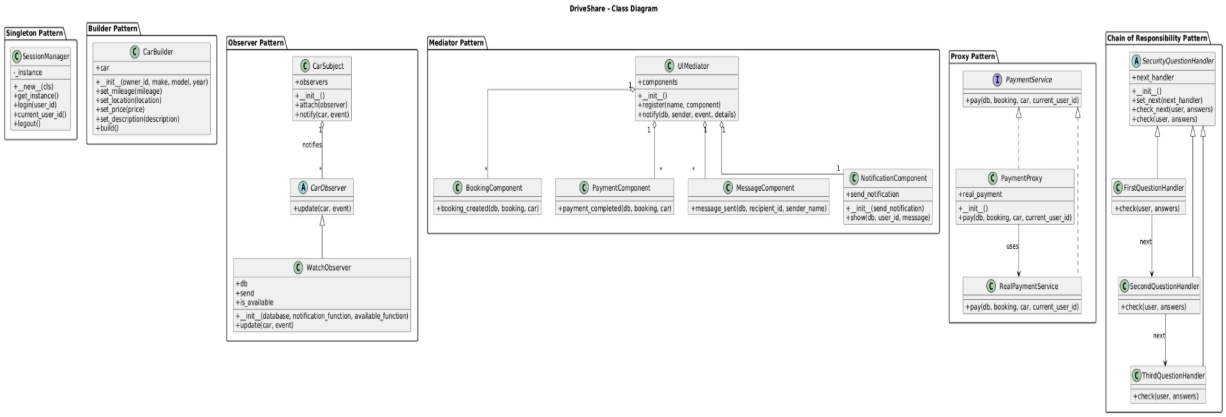
DriverShare Project
CIS 476
Maria Aldhaheri
11/9/2026

1. Class diagram



The class diagram shows the main classes used in DriverShare and how they connect. The project uses the six design patterns required in the assignment:

- Singleton - SessionManager: Uses one shared session manager for login, logout, and checking the current user.

- Builder - CarBuilder: Builds a car listing step by step, with methods for adding details such as mileage, location, price, and description.

- Observer - CarSubject and WatchObserver: Checks Renters' watchlist settings and sends an alert when a car's price or availability matches what they saved.

- Mediator - UIMediator: Passes information between the booking, payment, message, and notification components.

- Proxy - PaymentProxy: checks a demo payment before passing it to the payment service.
- C
- Chain of Responsibility - security question handlers: Checks the three recovery answers in order. If an answer is wrong, the password is not changed.

The pattern classes are located in the patterns folder. The class diagram should use the class names and connections from those Python files.

2. Database Schema

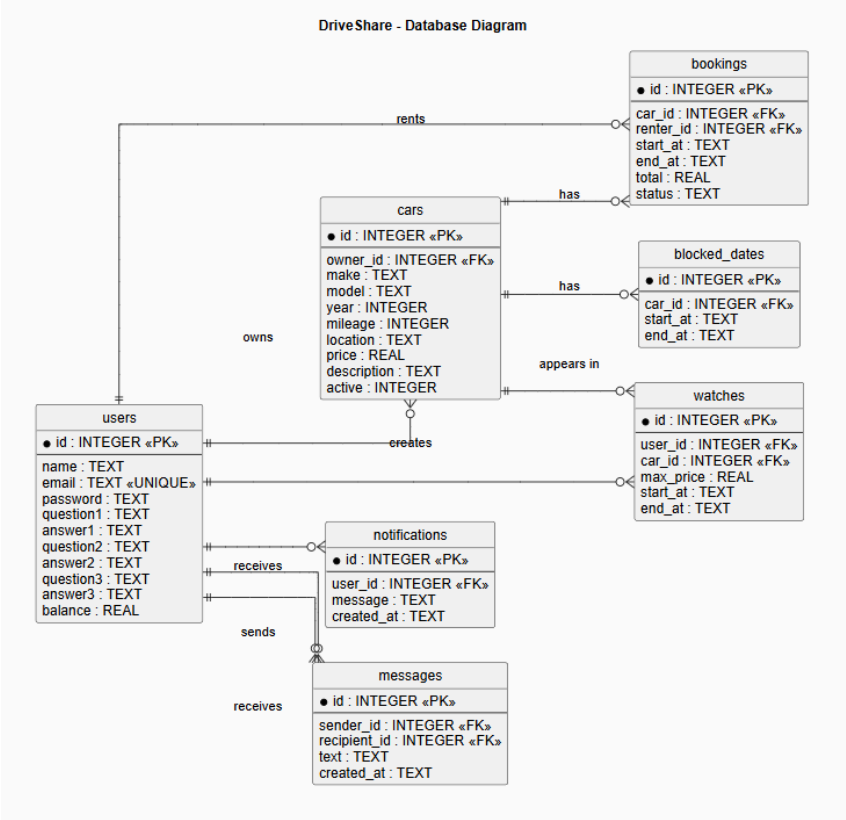
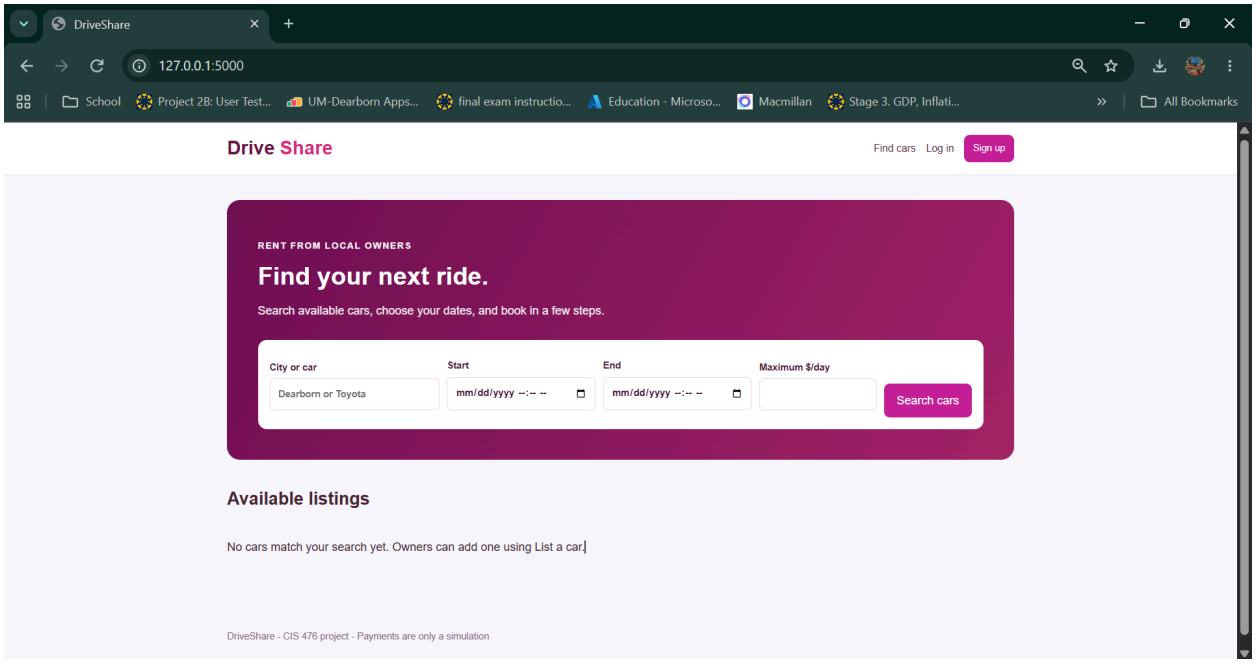


Table	Description
users	User account details, password hash, security questions and answer hashes, and demo balance
cars	Car details, owner, pickup location, price, description, and listing status
blocked_dates	Dates and times blocked by a car owner
bookings	Car, renter, rental dates, total price, and booking status
watches	Cars saved by renters, their price limit, and optional dates
notifications	Notifications saved for users
messages	Messages sent between users

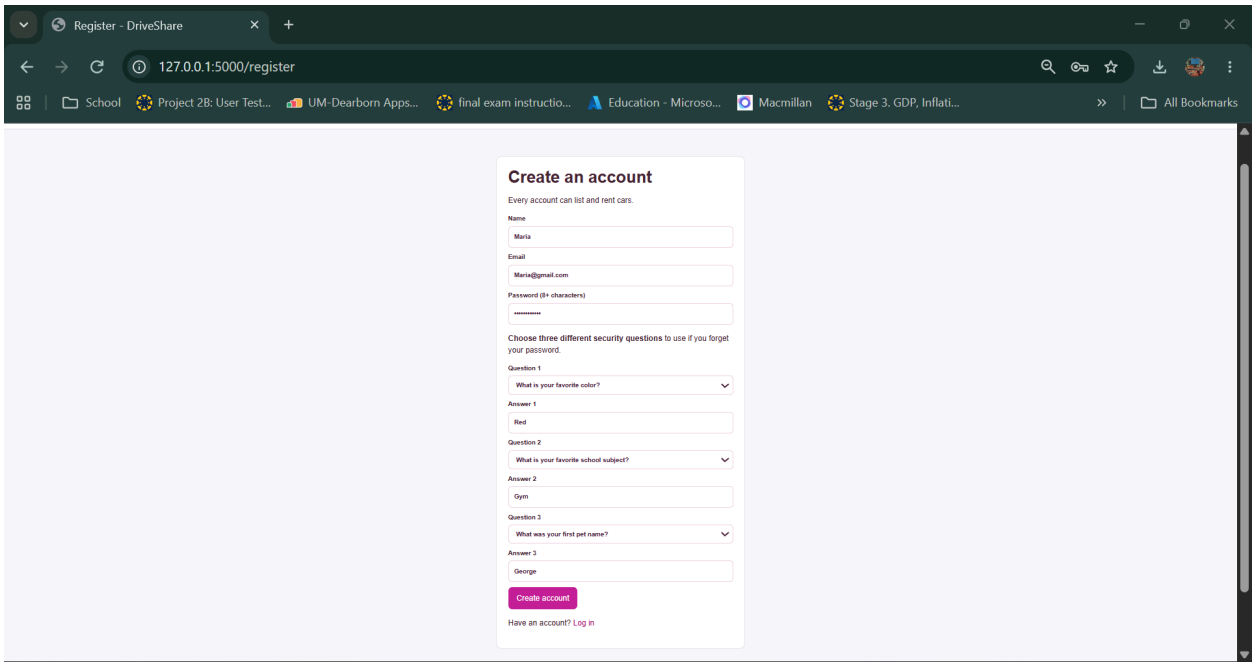
The Id field identifies each row in a table. Foreign keys connect related records. For example, a car is connected to its owner, while a booking is connected to a car and a renter.

3. User Interface Screenshots

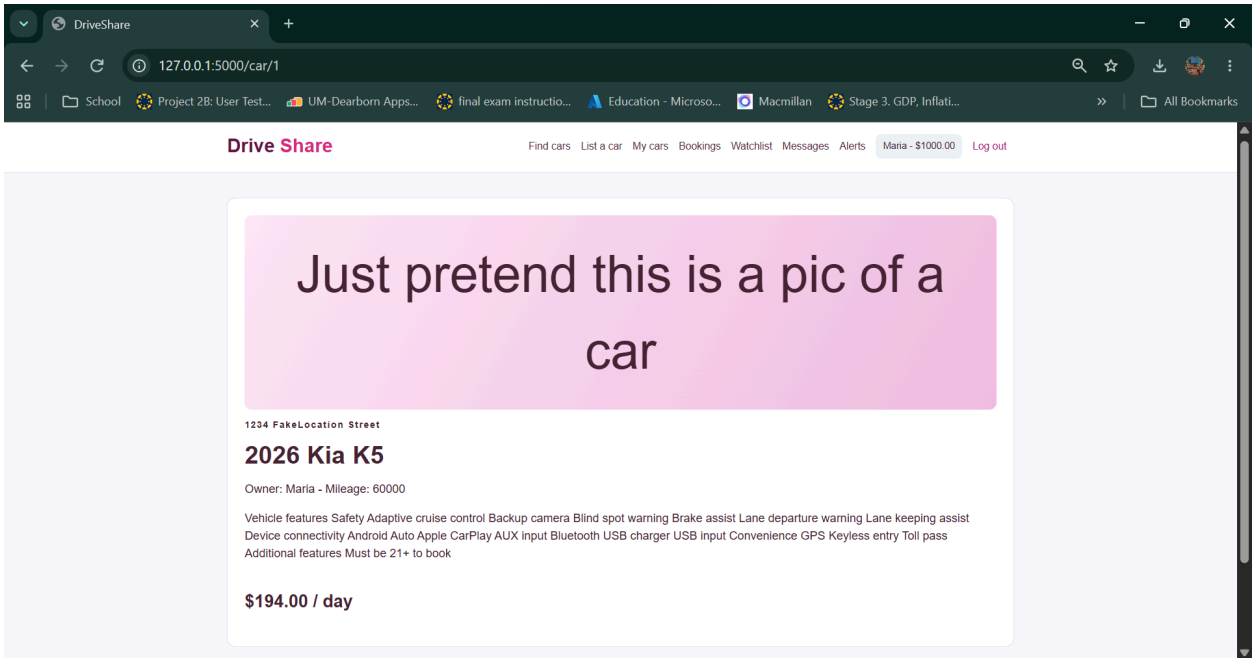
Homepage:



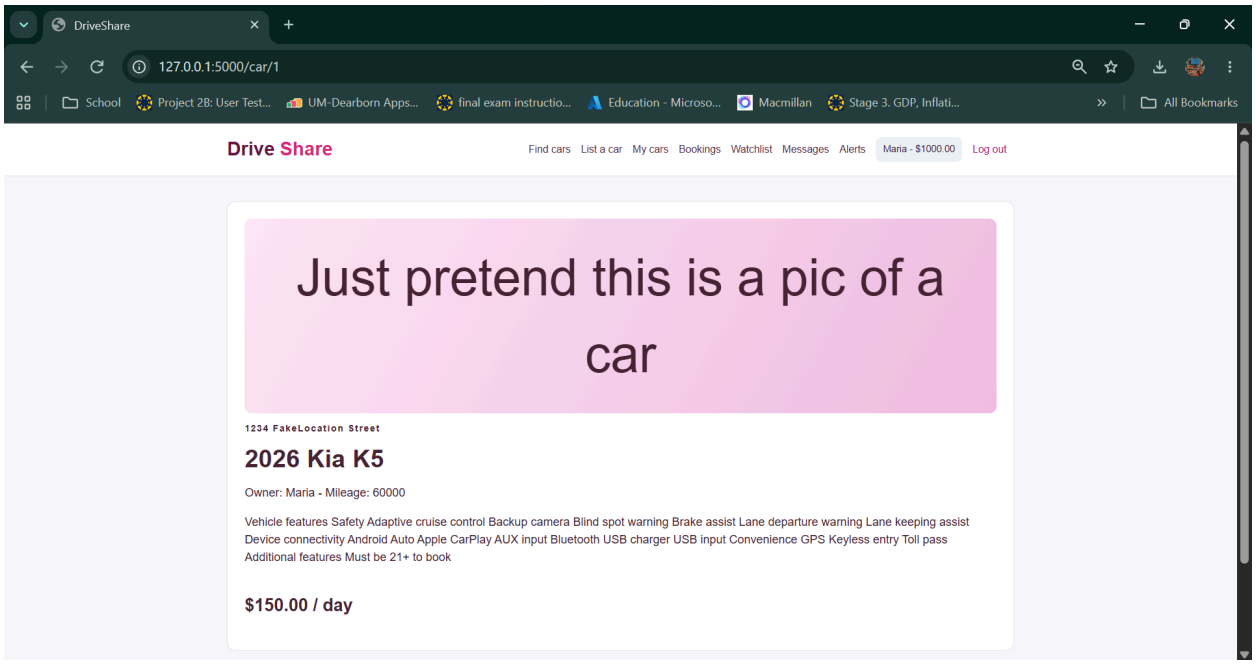
Registration Page:



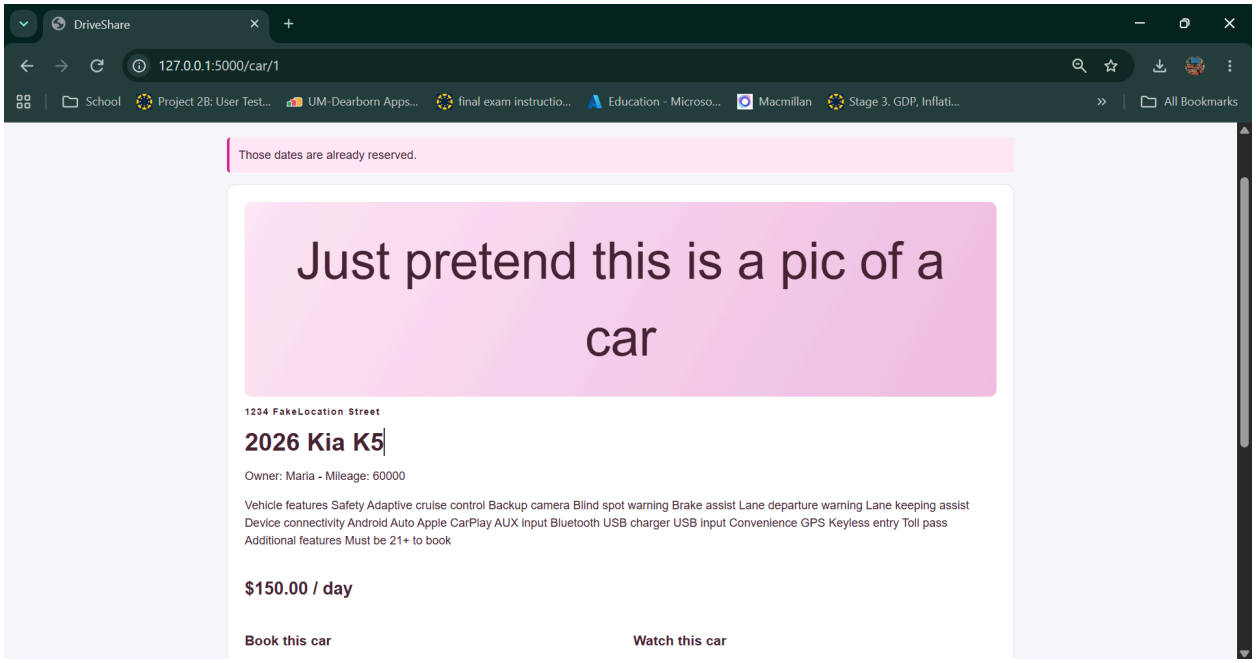
Car Details:



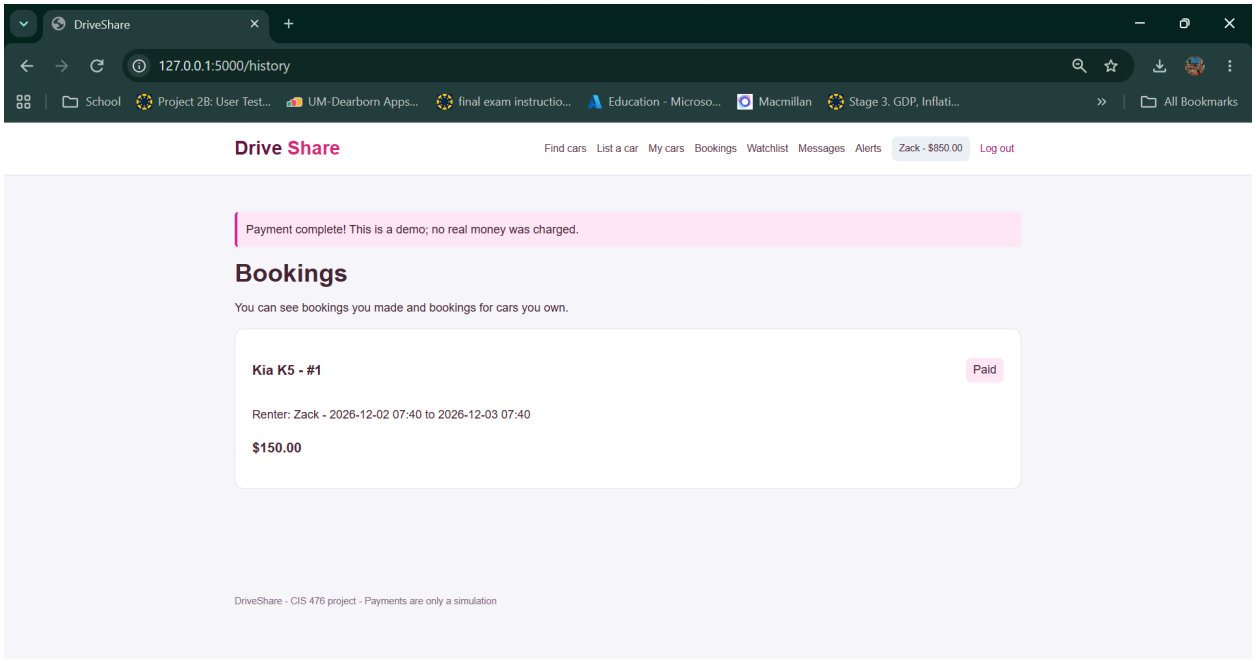
Edit Car Listing:



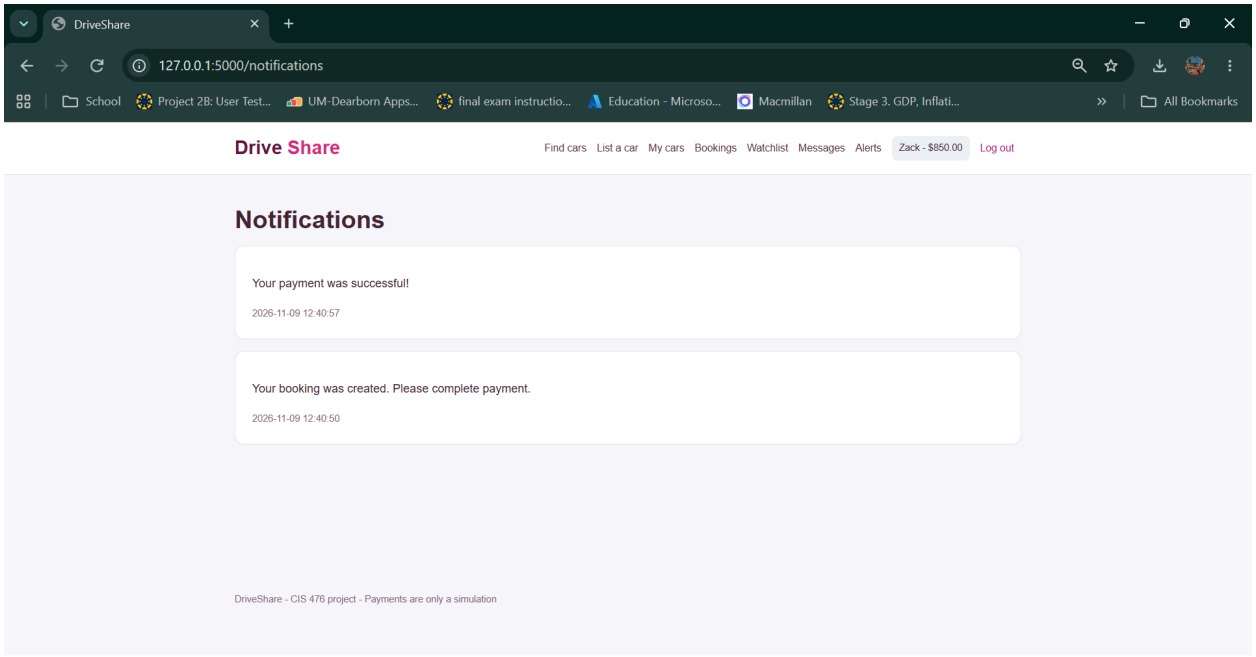
Booking Rejection-Available Dates:



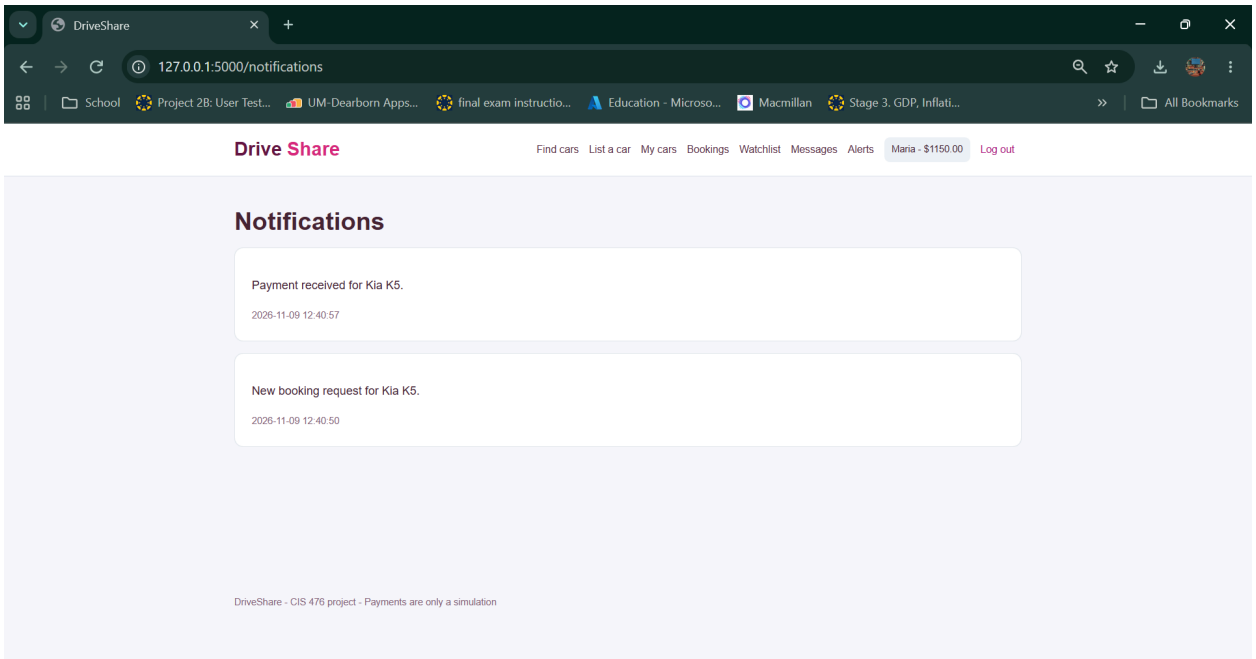
Completed Booking and Payment:



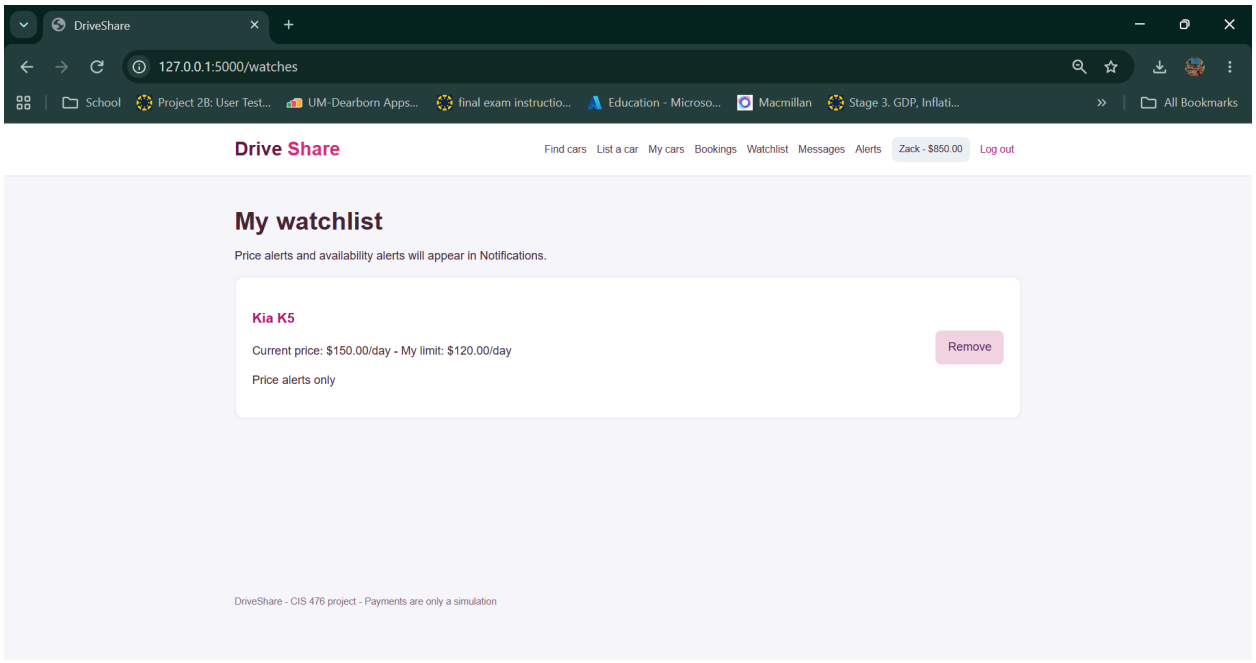
Renter Payment Notifications::



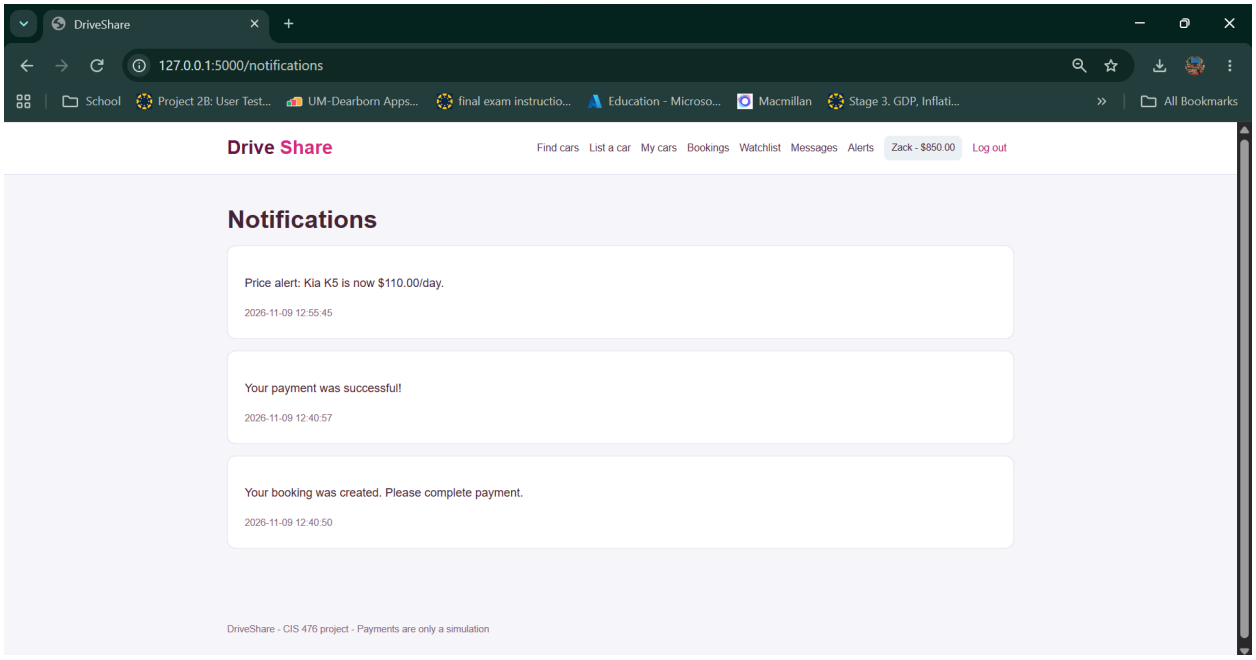
Owner Booking and Payment Notifications:



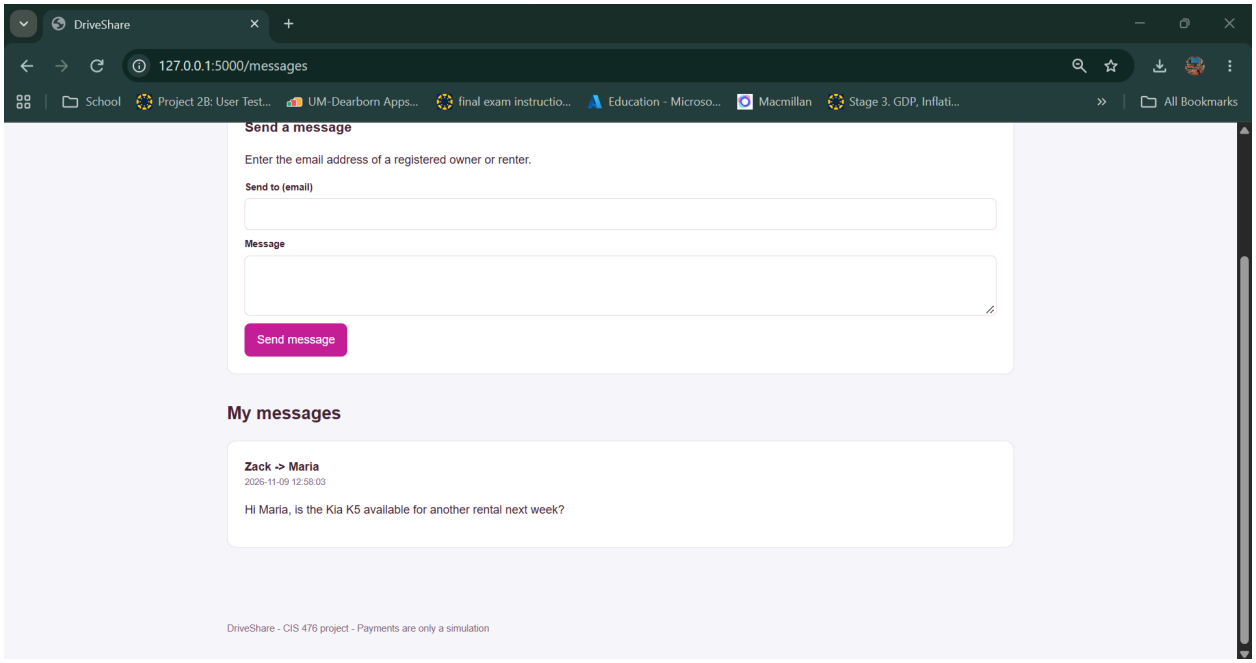
Renter Watchlist:



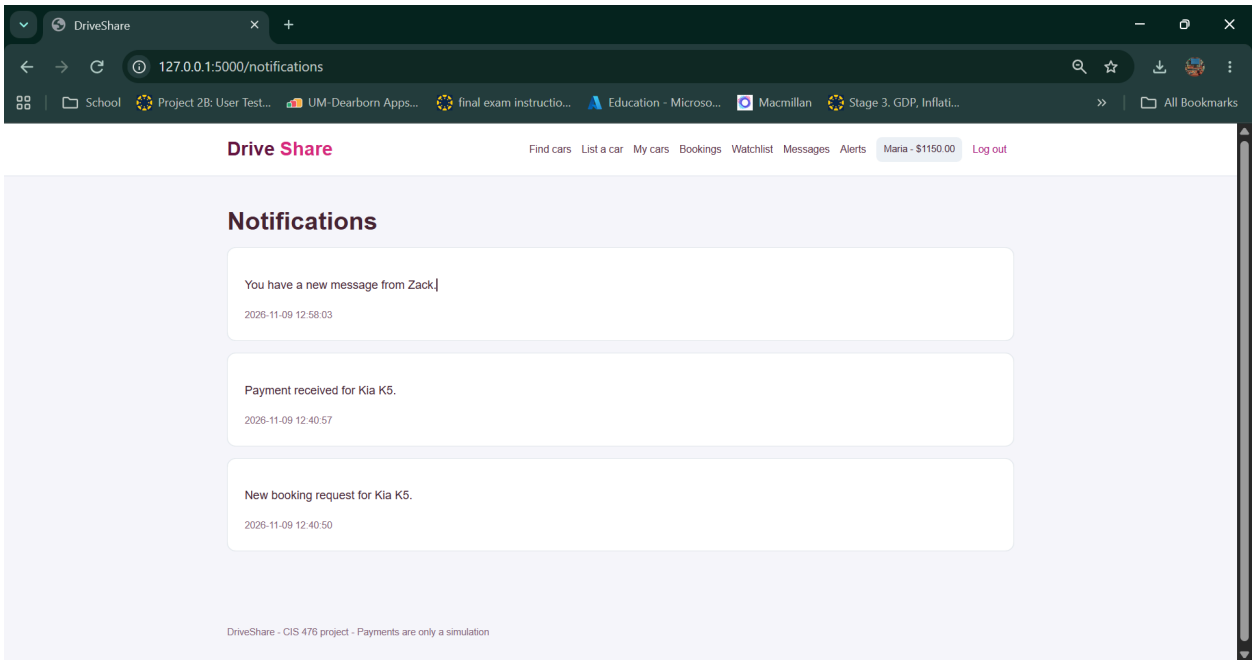
Automatic Price Alert:



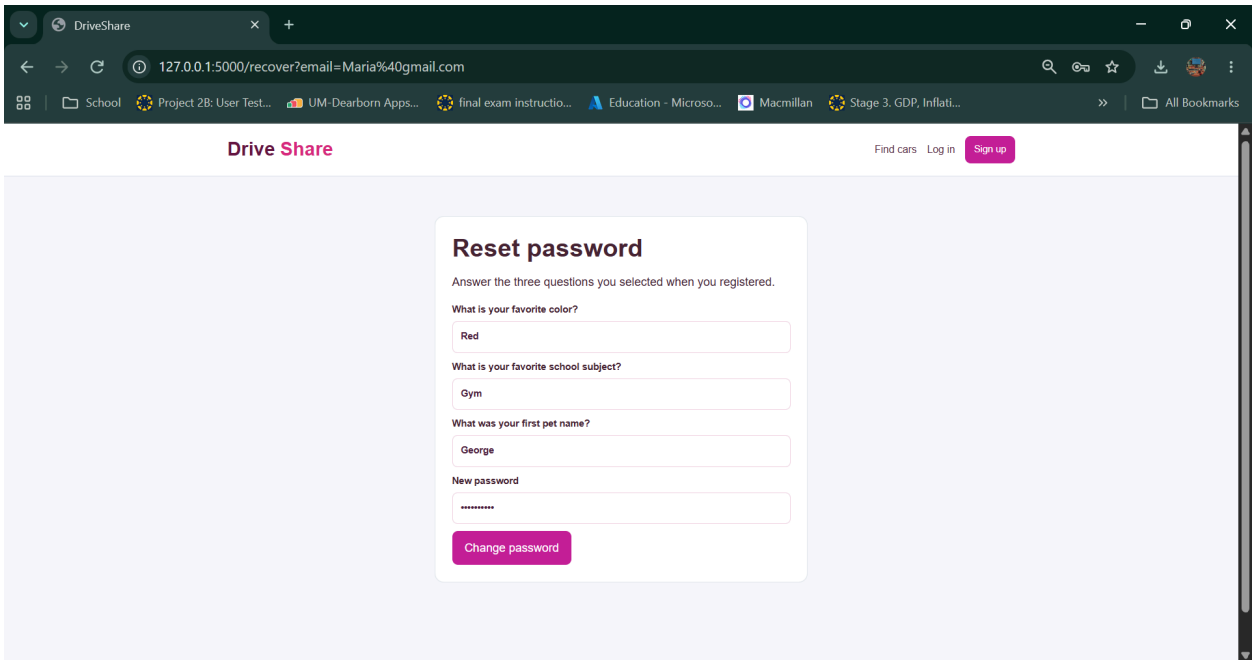
New Message Sent:



New Message Notification:



Password Recovery:



Rejected Password Recovery:

